



SOURCE SYSTEM ANALYSIS & ERD

Document 02 of 09 · B2B Data Engineering Platform



Documents the analysis of the source systems: entities, relationships, business rules, data profiling and data-quality findings.

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Document Control

Revision History

Version	Date	Author	Description of Change
0.1	-	Syed Kazim Raza	Template created
1.0	28/09/2026	Syed Kazim Raza	Content populated from source DDL and read-only profiling of SQL Server and CSV files
1.1	30/09/2026	Syed Kazim Raza	Profiling re-run on the rebuilt data; dirty CSV rows (DQ-04)
1.2	04/10/2026	Syed Kazim Raza	Source ERD redrawn (light theme, editable draw.io source in docs/diagrams)
1.3	05/10/2026	Syed Kazim Raza	Formatting to documentation standard: table column widths, numbered figure and table captions with lists, figure descriptions, complete acronym list

Approvals

Role	Name	Signature	Date
Prepared by	Syed Kazim Raza		
Reviewed by	[Team Lead]		
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Related Documents

Ref	Document	Location
R1	Business Requirements Document (BRD)	Provided by DataRopes
R2	Project Document (technical reference)	docs/deliverables/project_document.md
R3	Project Plan	docs/deliverables/project_plan.md
R4	README / Deployment Guide	README.md

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1. Introduction

1.1 Purpose

This document records what was learned about the source systems before modelling and building the pipelines.

1.2 Scope

In scope:

- SQL Server b2b_source_db (8 tables) and the marketing-leads and web-logs CSV feeds
- Entity relationships, source business rules, profiling and data quality

Out of scope:

- Target data models (see LDM / PDM)
- Interface mechanics (see ISD)

1.3 Intended Audience

Table 1: Intended Audience

Audience	How they use this document
Data Engineers / Modellers	Understand the sources before modelling
Team Lead	Validate that the analysis is complete

1.4 Definitions & Acronyms

Table 2: Definitions & Acronyms

Term	Definition
B2B	Business-to-Business: companies selling to other companies.
BR	Business Rule: a rule the data must follow, numbered BR-1 ... in the Source System Analysis.
BRD	Business Requirements Document: here the Project Requirements Document provided by DataRopes.
Business key	Source identifier of an entity: 32-character hex UUID generated in Python (no IDENTITY columns).
CDC	Change Data Capture: how new and changed records can be identified in a source.
CSV	Comma-Separated Values: the text-file format of the marketing leads and web logs feeds.
DDL	Data Definition Language: SQL that creates schemas, tables, indexes and views.
DEP	Data Engineering Platform: prefix of the document IDs (B2B-DEP-01 ...).
DQ	Data Quality: DQ-x numbers the data-quality findings of the source analysis.
Entity	A business object stored as one source table or file (e.g. Orders).
ERD	Entity-Relationship Diagram: a diagram of tables (entities) and their relationships.
FK	Foreign Key: a column that references the primary key of another table.
HTTP	Hypertext Transfer Protocol: the web protocol recorded in the web logs (GET, POST ...).
ID	Identifier.
ISD	Interface Specification Document (deliverable 01).
KPI	Key Performance Indicator: a business measure shown on the dashboard (KPI-01 ... KPI-15).
LDM	Logical Data Model (deliverable 03).
Load level	Dependency level 0-4: an entity can only be loaded after the entities it references.
OLTP	Online Transaction Processing: an operational database such as the SQL Server source.
PDM	Physical Data Model (deliverable 04).

Term	Definition
Profiling	Measuring row counts, nulls, distinct values and ranges to understand data content and quality.
SCD	Slowly Changing Dimension: SCD1 overwrites changes, SCD2 keeps every version with validity dates.
SQL	Structured Query Language.
UQ	Unique constraint: no two rows may have the same value.
UTF-8	Unicode character encoding used by the CSV files.
UTM	Urchin Tracking Module: campaign parameters (utm_source, utm_medium, utm_campaign).

2. Source Systems Overview

Table 3: Source Systems Overview

Source ID	Name	Type	Technology	Business Domain	Entities	Update Pattern
S1	b2b_source_db	Operational database (OLTP)	Microsoft SQL Server 16.0.1000.6 Express, schema source	Companies, customers, suppliers, catalogue, orders	8 tables	Inserts and updates, updated_at maintained, no deletes
S2	marketing_leads.csv	Flat file (CSV)	Local file, comma separated, UTF-8	Marketing / lead funnel	1 file	Cumulative file; new leads appended, changed leads rewritten in place
S3	web_logs.csv	Flat file (CSV)	Local file, comma separated, UTF-8	Web platform traffic	1 file	Append-only log

3. Source Entity–Relationship Diagram

Figure 1 shows the source data model: the eight SQL Server tables with their keys and relationships, plus the two CSV feeds.

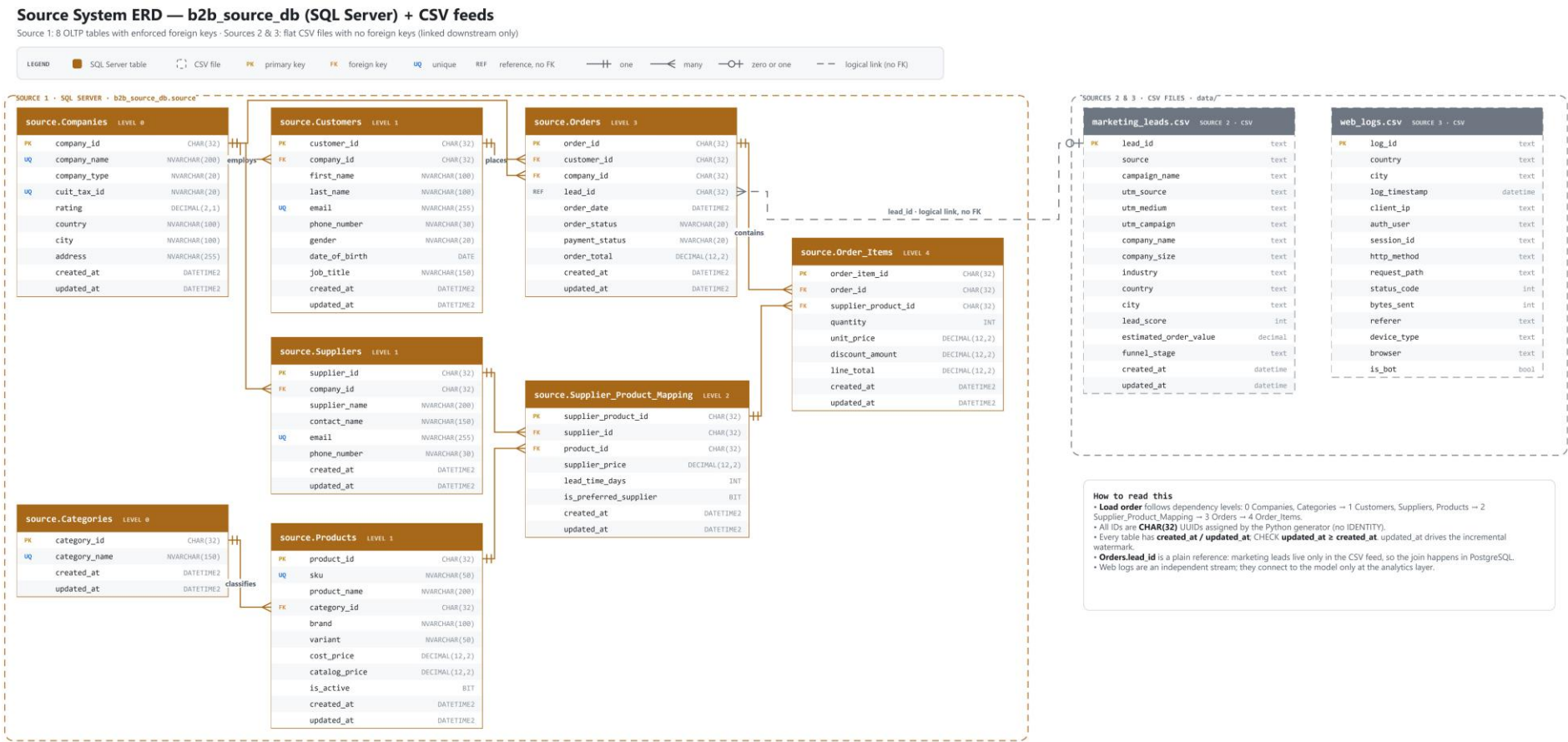


Figure 1: Source system ERD — b2b_source_db and CSV feeds

4. Entity Catalogue

Table 4: Entity Catalogue

Entity	Source	Description	Business Key	Load Level	Row Count
Companies	S1	Buyer and supplier organisations on the platform	company_id	Level 0	2,500
Categories	S1	Product category catalogue	category_id	Level 0	10
Customers	S1	Buyer contact persons who place orders	customer_id	Level 1	94,755
Suppliers	S1	Supplier accounts (one per supplier company)	supplier_id	Level 1	400
Products	S1	Products with cost and catalog price	product_id (UQ sku)	Level 1	96
Supplier_Product_Mapping	S1	Which supplier sells which product, price and lead time	supplier_product_id (UQ supplier_id + product_id)	Level 2	347
Orders	S1	Order header: customer, company, status, payment, total	order_id	Level 3	156,678
Order_Items	S1	Order lines: supplier-product, quantity, price, discount	order_item_id (UQ order_id + supplier_product_id)	Level 4	470,263
marketing_leads	S2	Marketing leads with campaign, UTM and funnel stage	lead_id	Level 0 (independent)	150,616
web_logs	S3	HTTP requests on the B2B portal	log_id	Level 0 (independent)	301,220

5. Relationships & Cardinality

Table 5: Relationships & Cardinality

Parent	Child	Join Column(s)	Cardinality	FK Enforced?	Notes
Companies	Customers	company_id	1 : 10-80 (buyer companies only; 2,100 have customers)	Yes (FK)	BR-1: customers belong only to Buyer companies
Companies	Suppliers	company_id	1 : 1 (400 supplier companies, 400 suppliers)	Yes (FK)	BR-2: suppliers belong only to Supplier companies
Categories	Products	category_id	1 : many (10 categories, 96 products)	Yes (FK)	BR-3: product belongs to one category
Suppliers	Supplier_Product_Mapping	supplier_id	1 : 0..many (234 of 400 suppliers have offers)	Yes (FK)	Resolves the many-to-many Supplier-Product

Parent	Child	Join Column(s)	Cardinality	FK Enforced?	Notes
Products	Supplier_Product_Mapping	product_id	1 : 2-5	Yes (FK)	BR-4: product has several suppliers; UNIQUE(supplier_id, product_id)
Customers	Orders	customer_id	1 : 0..many (57,396 customers have ordered)	Yes (FK)	BR-5
Companies	Orders	company_id	1 : 0..many	Yes (FK)	Must equal the customer's company (0 violations)
Orders	Order_Items	order_id	1 : 1..many (avg 3.0 lines)	Yes (FK)	BR-6; UNIQUE(order_id, supplier_product_id)
Supplier_Product_Mapping	Order_Items	supplier_product_id	1 : 0..many	Yes (FK)	An order line buys a specific supplier offer
marketing_leads (S2)	Orders	lead_id	1 : 0..1 (6,044 orders from leads)	No (cross-source)	BR-8; validated in the intermediate layer

6. Business Rules Observed in the Source

Table 6: Business Rules Observed in the Source

Rule ID	Rule	Entities	Enforced By
BR-1	Customers belong only to Buyer companies (10-80 customers each)	Companies, Customers	Generator; checked by profiling (0 violations)
BR-2	Suppliers belong only to Supplier companies	Companies, Suppliers	Generator; checked by profiling (0 violations)
BR-3	A product belongs to exactly one category	Products	FK + NOT NULL
BR-4	A product has 2-5 suppliers; one offer per supplier-product pair	Supplier_Product_Mapping	Generator; UNIQUE constraint
BR-5	An order belongs to one customer and cannot predate the customer	Orders, Customers	FK; generator (0 violations)
BR-6	An order has one or more order items	Orders, Order_Items	Generator (0 orders without items)
BR-7	order_total = SUM(line_total); line_total = quantity x unit_price - discount	Orders, Order_Items	Generator (0 mismatches)
BR-8	Every Won lead produces exactly one order; conversion derived from Orders.lead_id	marketing_leads, Orders	Generator; conversion derived from orders
BR-9	updated_at >= created_at	All tables	CHECK constraints
BR-10	Cancelled orders stay in the data but are excluded from revenue	Orders	Downstream KPI logic (7,613 cancelled orders)
BR-D1	Domains: company_type, gender, order_status, payment_status; rating 1.0-5.0; prices, amounts >= 0; catalog_price >= cost_price	Several	CHECK constraints

Rule ID	Rule	Entities	Enforced By
BR-D2	Unique company_name, cuit_tax_id, category_name, sku, customer and supplier e-mail	Several	UNIQUE constraints

7. Data Profiling Results

Profiling was run read-only against the SQL Server source and the two CSV files on 30/09/2026 (row counts, nulls, distinct values, ranges and cross-table checks), extending the queries in [sql_server/exploration/dataset_validations.sql](#). The pipeline also prints shape, dtypes and null counts of every extracted batch. Key columns:

Table 7: Data Profiling Results

Table	Column	Rows	Null %	Distinct	Min	Max	Issues
Companies	company_id	2,500	0.0%	2,500	-	-	Unique
Companies	Rating	2,500	0.0%	26	2.5	5.0	Range 2.5-5.0 within 1.0-5.0
Customers	company_id	94,755	0.0%	2,100	-	-	2,100 buyer companies
Customers	phone_number	94,755	0.0%	91,307	len 12	len 17	Digits only
Products	catalog_price	96	0.0%	96	3.60	20547.27	17 products inactive
Supplier_Product_Mapping	lead_time_days	347	0.0%	19	3	21	3-21 days
Orders	order_date	156,678	0.0%	156,298	2024-09-30 10:07:45	2026-09-30 17:27:15	~2 years of history
Orders	lead_id	156,678	96.1%	6,044	-	-	Null = not from a lead (expected)
Orders	order_total	156,678	0.0%	154,123	2.40	303352.99	0 mismatches with lines
Order_Items	Quantity	470,263	0.0%	10	1	10	1-10 units
Order_Items	discount_amount	470,263	0.0%	192,789	0.00	27899.36	0 line_total mismatches
marketing_leads.csv	lead_score	150,616	0.0%	102	0	150	0 duplicate lead_id; 723 outside 1-100 (DQ-04)
marketing_leads.csv	estimated_order_value	150,616	0.0%	148,515	-1000.00	1999961.38	741 negative (DQ-04)
web_logs.csv	auth_user	301,220	5.0%	158,330	3	22	Anonymous requests
web_logs.csv	session_id	301,220	0.0%	301,220	32	32	One id per request
web_logs.csv	status_code	301,220	0.0%	9	200	999	8 valid codes + 999 (DQ-04)

Figure 2 shows the row count of every source table in SQL Server at the time of profiling.

```
=====
2. Row Count Exploration
=====
*/

SELECT 'Companies' AS table_name, COUNT(*) AS row_count
FROM source.Companies

UNION ALL

SELECT 'Customers', COUNT(*)
FROM source.Customers

UNION ALL

SELECT 'Categories', COUNT(*)
FROM source.Suppliers

UNION ALL

SELECT 'Products', COUNT(*)
FROM source.Products

UNION ALL

SELECT 'Suppliers', COUNT(*)
FROM source.Suppliers

UNION ALL

SELECT 'Supplier_Product_Mapping', COUNT(*)
FROM source.Supplier_Product_Mapping

UNION ALL

SELECT 'Orders', COUNT(*)
FROM source.Orders

UNION ALL

SELECT 'Order_Items', COUNT(*)
FROM source.Order_Items;
```

100 %

Results Messages

	table_name	row_count
1	Companies	2500
2	Customers	94755
3	Categories	400
4	Products	96
5	Suppliers	400
6	Supplier_Product_Mapping	347
7	Orders	156883
8	Order_Items	470867

Figure 2: Row counts of the source tables in SQL Server

Figure 3 shows the first rows of the marketing leads file as delivered, with its header row and comma-separated layout.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	lead_id	source	campaign_name	utm_source	utm_medium	utm_campaign	company_name	company_size	industry	country	city	lead_score	estimated_order_value	funnel_stage	created_at	updated_at
2	3ecb55e90827474a8623121de0bbf37	Cold Call	Summer Sale	linkedin	social	summer_sale	Beck-Walsh	1-Oct	Technology	United States	Seattle	7	5747.47	New	21/9/2025 5:02	4/5/2026 6:15
3	e13740d546f1417099300965c1313063	Website	Industry Expo	facebook	referral	industry_expo	Alexander, Johnson and Kemp	201-500	Education	United Kingdom	London	15	189761.32	New	9/3/2025 6:20	16/6/2025 23:22
4	b505d6e19e9c4914b3b52c97a42a03f	Website	Referral Drive	linkedin	email	referral_drive	Horn-Brown	Nov-50	Retail	Australia	Sydney	20	39506.8	Contacted	8/1/2025 21:01	23/2/2025 22:37
5	5336f85ef7314abf801e4da81d15230f	Referral	Product Launch	newsletter	cpc	product_launch	Harrington Ltd	Nov-50	Logistics	United States	Seattle	42	27336.7	Qualified	7/6/2026 12:07	7/8/2026 7:47
6	684a78f718fa4906b9d00763e21ca72	Email Campaign	Enterprise Growth	linkedin	organic	enterprise_growth	Kirby LLC	1-Oct	Retail	United States	Seattle	33	8569.17	Qualified	8/7/2026 17:05	28/8/2026 12:40
7	2760dfe5bcd7430fba2172d255ffa25e	Trade Show	Holiday Promotion	linkedin	referral	holiday_promotion	Martin-Parker	51-200	Hospitality	United States	New York	43	148834.24	Lost	20/4/2026 17:43	14/9/2026 15:10
8	28714edfd314d7d88d533807639e85f	Partner	Referral Drive	bing	organic	referral_drive	Clark, Hamilton and Brown	1-Oct	Construction	United States	New York	13	2706.87	New	3/7/2026 18:22	1/8/2026 1:22
9	024a88bd95d744da8b3a67989bf991f	Partner	Free Trial Campaign	facebook	cpc	free_trial_campaign	Mcdonald LLC	Nov-50	Finance	United States	Austin	72	22210.9	Negotiation	27/12/2025 9:20	16/3/2026 23:56
10	7425579e761a420abdab60b0d74d0b	Website	Enterprise Growth	facebook	cpc	enterprise_growth	Ruiz Ltd	51-200	Technology	United States	New York	54	130368.2	Proposal	26/12/2024 4:46	23/5/2025 15:26
11	48f74b1a740342c6ad9fa249f2c978b	Email Campaign	Industry Expo	facebook	organic	industry_expo	Gaines, Mcneil and Wright	Nov-50	Construction	United States	Austin	32	47865.4	Contacted	9/8/2026 5:34	10/9/2026 10:40
12	2fd55742628746ae9b72ccad260f853f	Website	Industry Expo	linkedin	email	industry_expo	Griffin, Randall and Perry	51-200	Construction	United States	Chicago	74	115191.51	Negotiation	28/10/2024 20:14	16/7/2026 17:31
13	e8eba2a584f8428abaae04a36b0910c	Cold Call	Free Trial Campaign	newsletter	organic	free_trial_campaign	Bullock Group	Nov-50	Energy	United Kingdom	Manchester	41	47042.67	Qualified	30/12/2025 2:37	6/1/2026 14:37
14	f5798cc4e26477a8dbad5c2bda20c	Email Campaign	Free Trial Campaign	newsletter	referral	free_trial_campaign	Lambert-Thompson	1-Oct	Technology	United States	New York	20	3748.18	Contacted	30/5/2026 7:30	6/7/2026 22:11
15	3dee0eb80eb94a50bfc9e24cc5d648f6	Partner	Enterprise Growth	linkedin	cpc	enterprise_growth	Brown PLC	1-Oct	Education	United States	New York	49	9180.49	Proposal	19/6/2025 0:47	23/10/2025 20:00

Figure 3: Sample rows of marketing_leads.csv

Figure 4 shows the first rows of the web logs file as delivered.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	log_id	country	city	log_timestamp	client_ip	auth_user	session_id	http_method	request_path	status_code	bytes_sent	referrer	device_type	browser	is_bot
2	c4896ec91b654828a96b4bcc101eb26	United Kingdom	Manchester	26/2/2026 13:18	10.19.246.214	martyn70	f831053ec82048b09d43ffb9b3168c59	DELETE	/api/orders	200	356429	https://twitter.com	Desktop	Firefox	FALSE
3	f102f7333a9a4c329e36194897e4aeb6	United States	New York	5/3/2025 7:04	10.19.246.214	joshua70	304f5c647d0d43a98dde90cb0443f5cc	DELETE	/api/customers	200	191499	https://twitter.com	Mobile	Edge	FALSE
4	53fed5ce0d4d4591b7d9dcf78732dc67	United Kingdom	London	9/8/2026 8:52	101.221.232.233	bethany29	8001a630e3b0471da202422a6925c172	DELETE	/api/customers	400	854	https://bing.com	Desktop	Edge	FALSE
5	90012112126444f2948b514b9b6df45a	United Kingdom	Birmingham	29/5/2026 10:48	166.40.114.106	brianhayward	e9d717b039ad4ff2884591741d4430f1	PUT	/checkout	403	3609	https://twitter.com	Desktop	Chrome	FALSE
6	c3da80ca65ad433fbb20bcb207af9d94	United States	Seattle	6/3/2025 15:31	101.221.232.233	steven29	7c2d551a5dee4edfb2bde2896160534d	GET	/api/customers	200	433619	https://bing.com	Mobile	Chrome	FALSE
7	a373826495c144d1a50a41c10a518b19	United States	Seattle	1/5/2025 9:18	110.40.114.106	chadjohnson	b3dee6ff12504ae18993f48d68327b3d	POST	/checkout	200	123227	https://twitter.com	Desktop	Chrome	FALSE
8	2e72cdd13e5a438bb8b7b04b257c8713	Australia	Sydney	4/2/2025 12:23	10.19.246.214	joshua70	53169ea0498445ea9d10417e16a69bec	POST	/products/123	200	290857	https://google.com	Desktop	Chrome	FALSE
9	ebdd36c7c4564deb96ca17832a74eeba	United States	Austin	13/10/2025 13:40	200.42.216.22	owenkatherine	39c02eacfa472caa1f77f7d83a49033	GET	/products/123	301	474996	https://bing.com	Mobile	Chrome	FALSE
10	cf18343ebc534032b4aca41ebef0fdd1	United States	Seattle	2/3/2025 15:57	171.108.1.47	brendajohnson	d8490c7e4a504d199ef80ca76885860e	PUT	/api/customers	400	4267	https://twitter.com	Desktop	Safari	FALSE
11	90f6b1e554c34661a605461d14c20b4b	United Kingdom	London	10/7/2025 14:26	200.42.216.22	robertstracey	f0250e4db7b340248133e469b48825a1	PUT	/checkout	200	333133	https://twitter.com	Desktop	Chrome	FALSE
12	d35520bd477243d690d4638ead0a6f5a	United States	Dallas	3/8/2025 9:20	27.113.139.219	harrisvictoria	502145f3ccce4e71add490bb27fc7840	DELETE	/checkout	200	224653	https://twitter.com	Tablet	Chrome	FALSE
13	a4c8e058f3124d5f83ae75054fa0506a	United States	Dallas	20/4/2025 21:24	222.22.6.30	cruzchristopher	1b289edb85864a87ab529576e064c65c	DELETE	/login	200	169383	https://twitter.com	Desktop	Chrome	FALSE
14	2175990f5e9640f0a423db332d5f3b90	United States	New York	13/6/2026 14:16	4.236.116.216	ashleymartinez	2255a98876b94b328339afefc5b51397	GET	/login	404	4946	https://google.com	Desktop	Safari	FALSE
15	a2d4aa1da3ed46b4875ea2c6e3db75a4	United States	Chicago	25/7/2026 22:29	81.26.16.200	danielmaldonado	d03507eb0a5846aabc52ec584d0d7d4	PUT	/search?q=laptop	200	79354	https://bing.com	Mobile	Firefox	FALSE

Figure 4: Sample rows of web_logs.csv

8. Data Quality Findings

Table 8: Data Quality Findings

Finding ID	Table / Column	Issue	Impact	Resolution in Pipeline
DQ-01	All SQL Server tables	No quality violations found: 0 orphans, 0 role mismatches, 0 total/line mismatches, 0 orders before customer creation	Low - source constraints and generator enforce the rules	Rules re-validated in intermediate anyway (defence in depth)
DQ-02	Orders.created_at	created_at of all orders is the insert time (from 2026-09-30), while order_date covers 2024-09 to 2026-09	Medium - created_at is not the business date	Warehouse uses order_date for date_key and partitioning, never created_at
DQ-03	web_logs.auth_user	15,014 rows (5.0%) empty	Low - anonymous traffic is expected	Kept as NULL (column optional)
DQ-04	marketing_leads.csv, web_logs.csv	About 1.5% of CSV rows are broken on purpose: 2,258 leads (score 0/150, negative value, empty industry) and 4,517 web logs (status 999, negative bytes, PATCH, empty browser)	Medium - invalid rows must not reach the warehouse	Caught in the intermediate layer and saved in intermediate.rejected_records (Quarantine table) with a reason

9. Change Data Capture Analysis

Table 9: Change Data Capture Analysis

Table	Change Indicator	Deletes Handled?	Extraction Strategy
Companies	updated_at (1-3 updated per day)	No	Watermark on updated_at; SCD2 in warehouse
Categories	updated_at (static)	No	Watermark on updated_at
Customers	updated_at (3-5 updated per day)	No	Watermark on updated_at
Suppliers	updated_at (static)	No	Watermark on updated_at

Table	Change Indicator	Deletes Handled?	Extraction Strategy
Products	updated_at (0-2 deactivated per day; soft delete via is_active)	Soft delete only	Watermark on updated_at
Supplier_Product_Mapping	updated_at (static)	No	Watermark on updated_at
Orders	updated_at (180-230 new + 5-10 status changes per day)	No	Watermark on updated_at
Order_Items	updated_at (insert only)	No	Watermark on updated_at
marketing_leads.csv	updated_at exists, but file rewritten in place	No	Full snapshot reload (TRUNCATE + APPEND)
web_logs.csv	None (append-only log)	No	Full snapshot reload (TRUNCATE + APPEND)

10. Assumptions & Open Questions

Table 10: Assumptions & Open Questions

ID	Type	Description	Owner	Status
A1	Assumption	No physical deletes occur in the source; products are soft-deleted with is_active = 0.	Data Engineering	Accepted
A2	Assumption	updated_at is set on every change (all generator UPDATE statements set it).	Data Engineering	Accepted
A3	Assumption	Data is synthetic (Python Faker generators), as specified in the requirements.	Data Engineering	Accepted